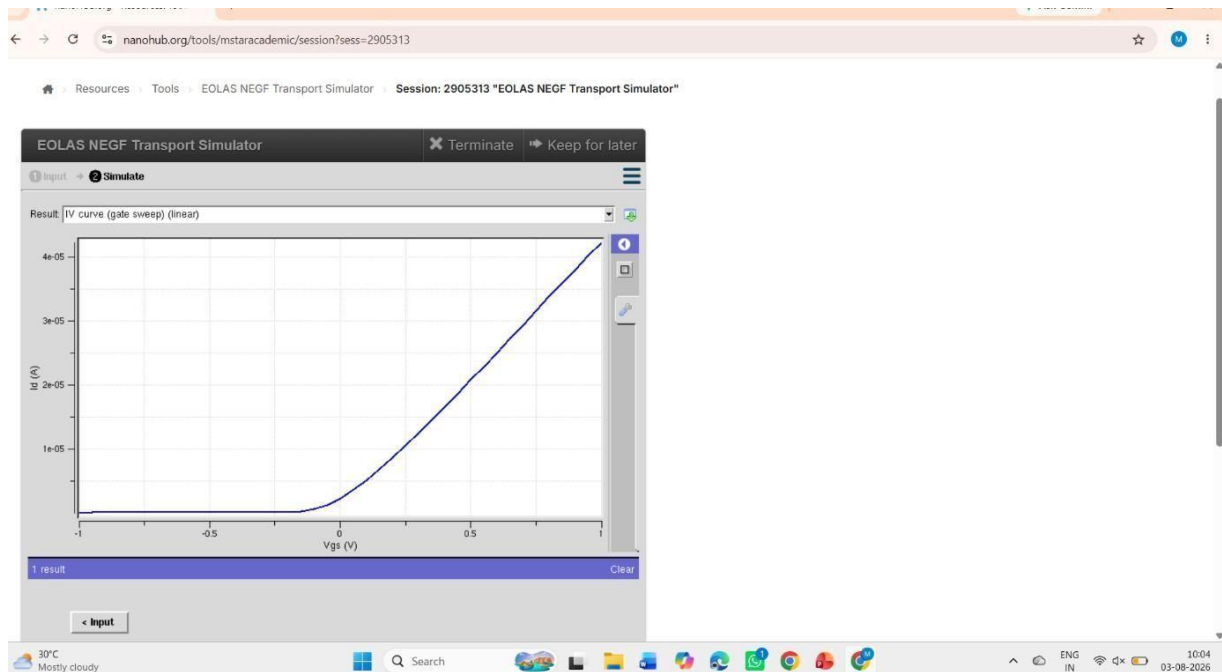


R. RUPASREE

192412256

Exp. 7. Simulation of VI Characteristics of FINFET, CNTFET and GFET using Nano Hub

(a) FINFET



Output: The output Drain current Performance Vs Gate supply voltage of the Fin FET device

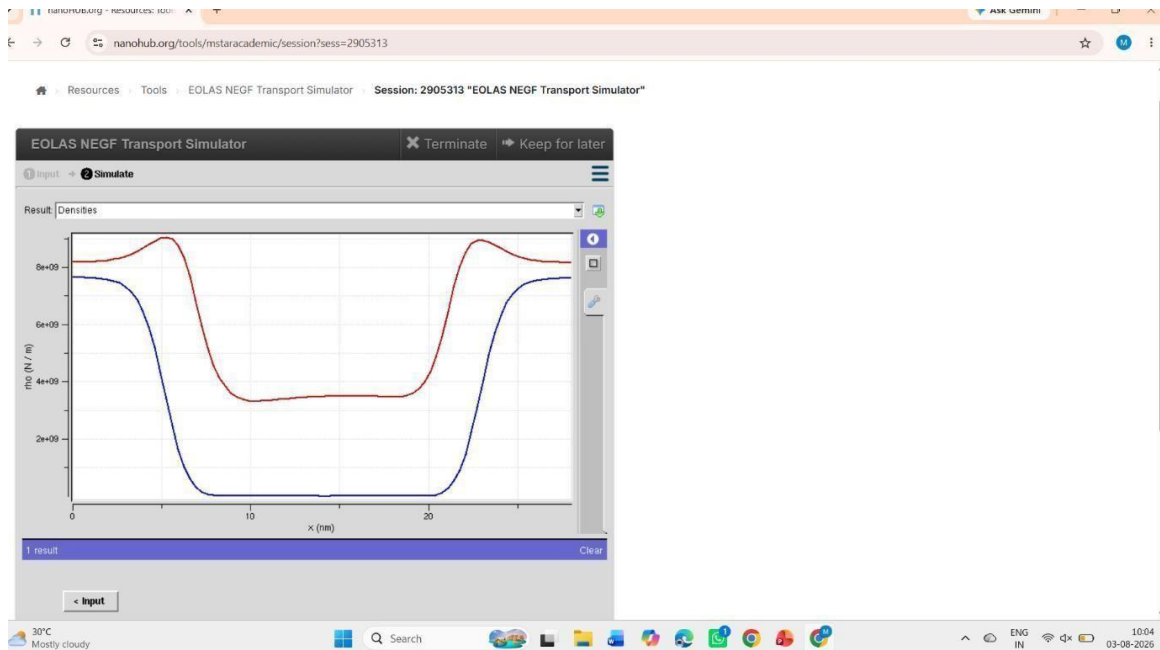
Outcome: The drain current (I_d) increases rapidly with increasing gate voltage (V_{gs}), demonstrating effective gate-controlled switching behaviour.

Case Study 1: Simulate & obtain the current spectrum characteristics for the last bias point of FinFET . Analyze the performance for different Oxide layer thicknesses



Outcome: The energy-resolved current is highest near the Fermi level and decreases with increasing energy, indicating carrier transport is concentrated at lower energies.

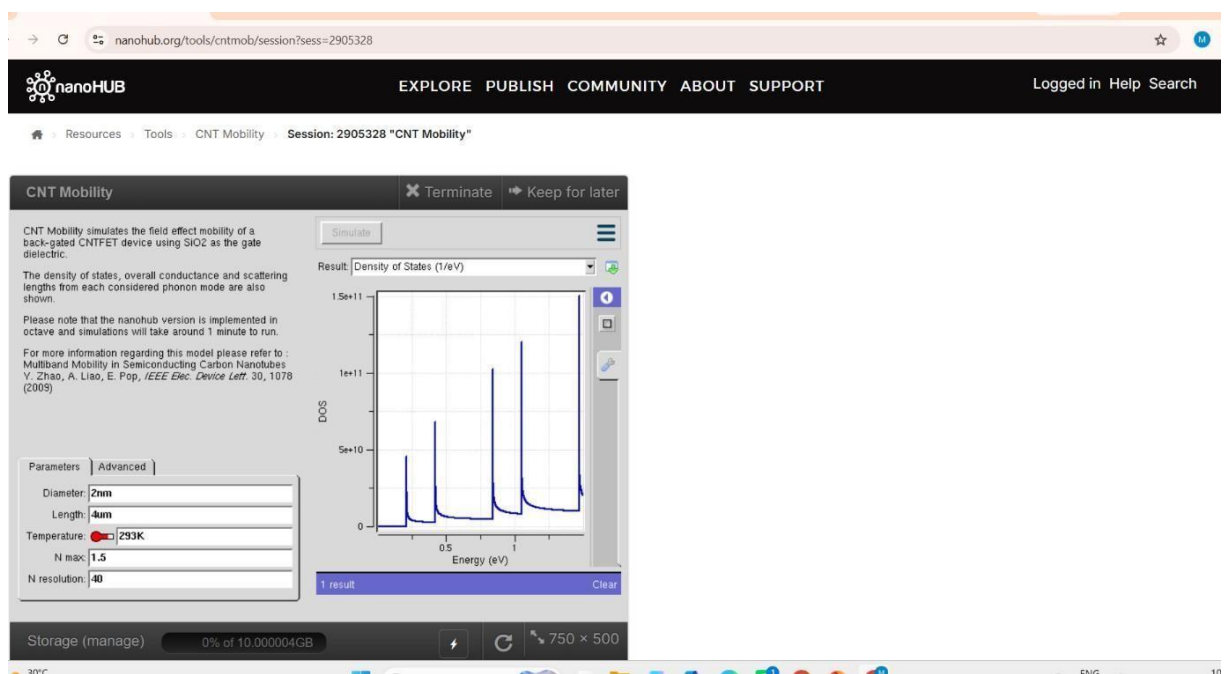
Case Study 2: Simulate & obtain the electron density per sheet characteristics of a FinFET. Analyse its performance with different Gate region lengths



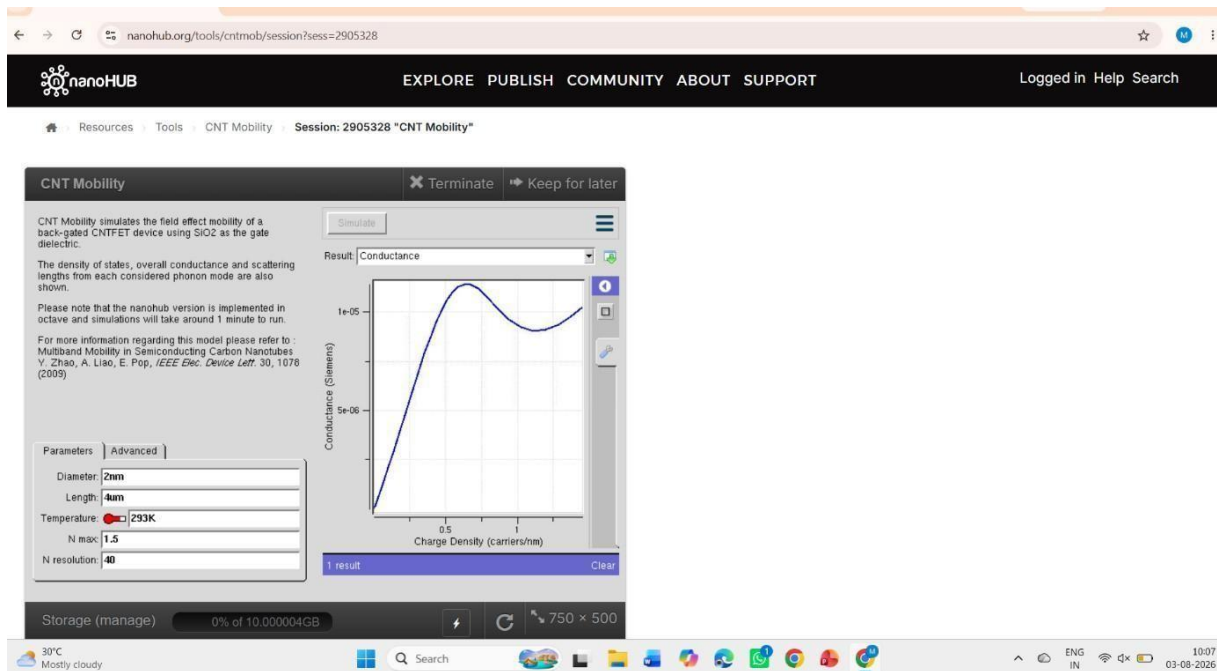
Outcome: The electron and hole densities vary across the device, showing carrier accumulation near the contacts and reduced density in the channel

(b) CNTFET

Output:

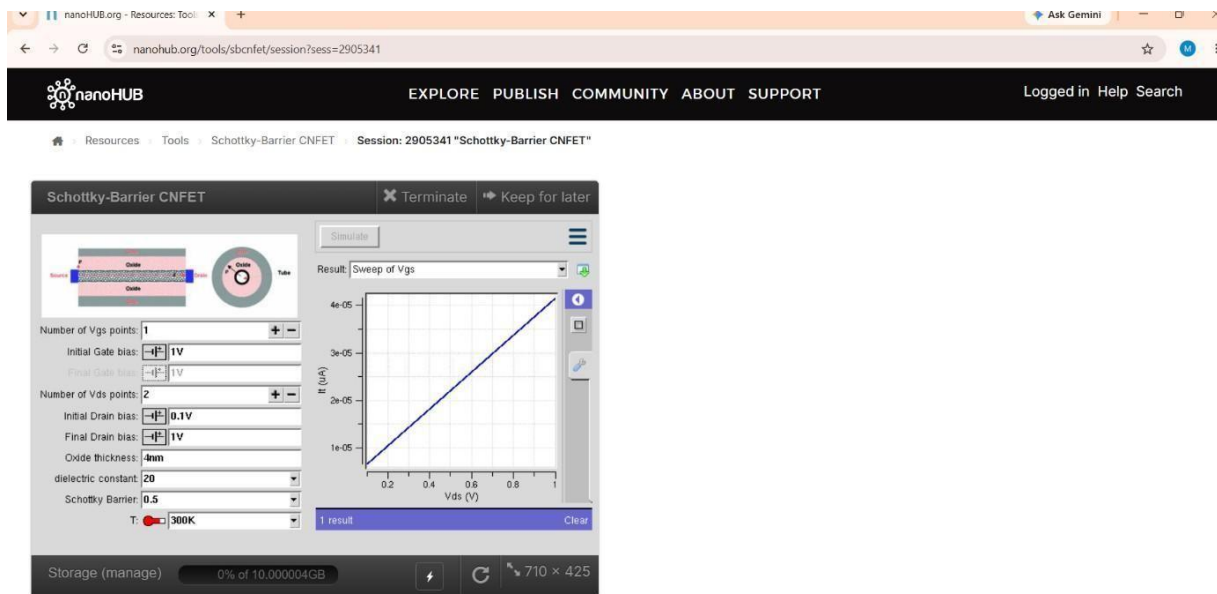


Outcome: The density of states exhibits sharp peaks with increasing energy, indicating quantized energy levels in the carbon nanotube.



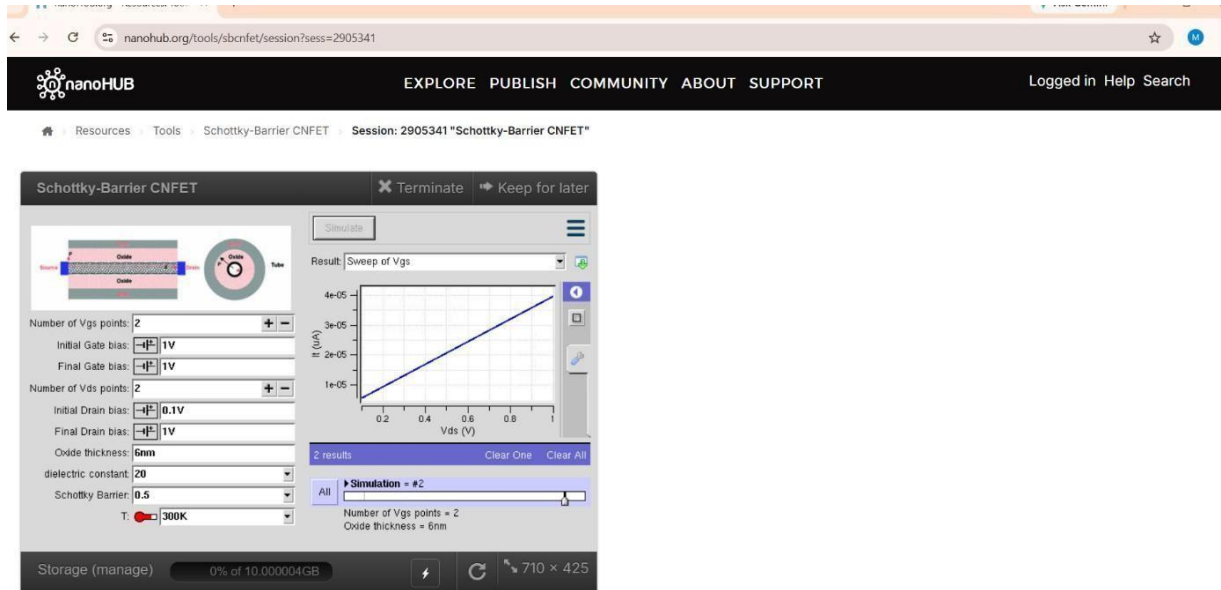
Outcome: The conductance increases with carrier density, showing improved electrical transport as more charge carriers are available.

Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness



Outcome: The drain current increases nearly linearly with drain voltage, indicating stable current transport through the Schottky-barrier CNFET.

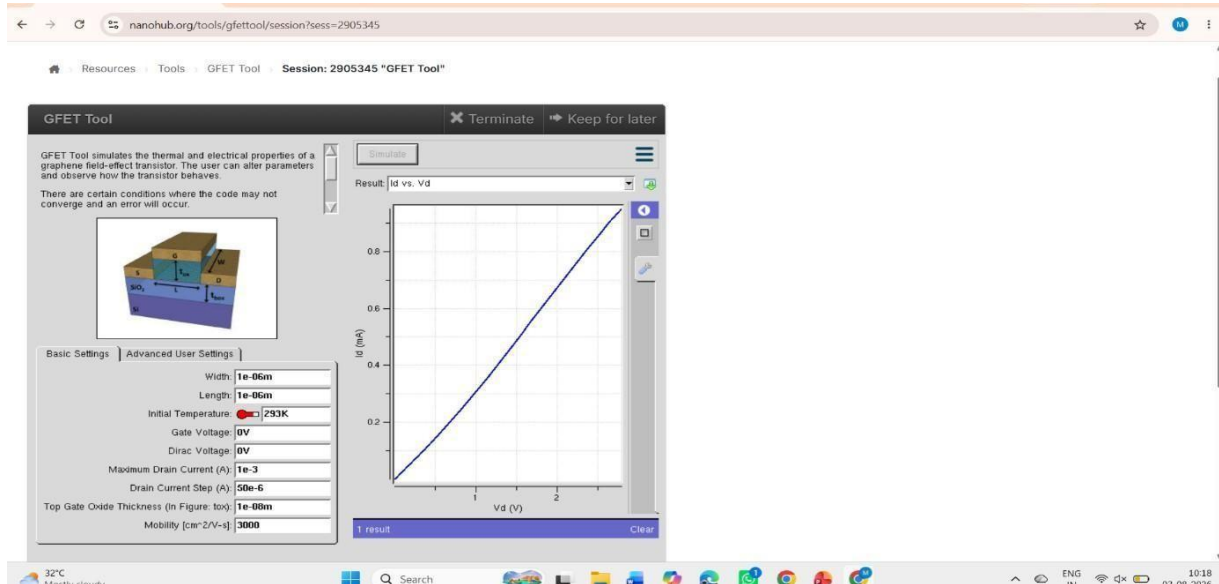
Case study 2: Simulate & Obtain the VI characteristics of CNT-FET by varying CNT diameter



Outcome: The drain current increases with applied drain voltage under different gate bias conditions, confirming effective transistor operation.

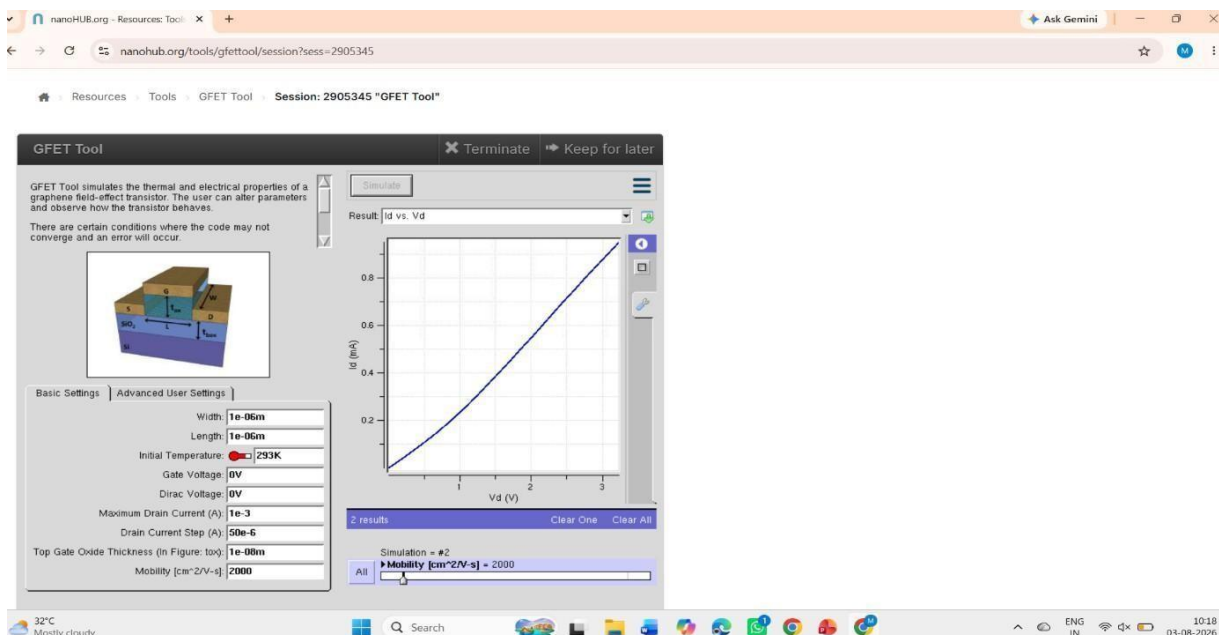
(c) GFET

Case Study 1: How does varying the gate voltage (V_{gs}) affect the drain current (I_{ds}) in a GFET?



Outcome: The drain current (I_d) increases almost linearly with drain voltage (V_d), indicating good current conduction in the GFET.

Case Study 2: What factors determine the maximum drain current ($I_{ds,max}$) in a Graphene Field Effect Transistor (GFET)



Output: The simulated VI curves show a relatively high drain current (I_{ds}) for a given gate-source voltage (V_{gs}), demonstrating efficient carrier transport in the graphene channel